Cryptography
— and its Types —

What is Cryptography?

Cryptography is the process of protecting information and communication by converting readable data (plaintext) into an unreadable form (ciphertext) so that unauthorized users cannot understand it.

It ensures:

- Data Confidentiality
- Data Integrity
- Authentication
- Non-repudiation

Simple Example

Original Message (Plaintext) → HELLO → After Encryption (Ciphertext) → XJ82@P

Only authorized users with the correct key can convert it back to the original message.

Objectives of Cryptography

01 Confidentiality – Protect data from unauthorized access

02 Integrity – Ensure data is not modified

03 Authentication – Verify the identity of users

04 Non-repudiation – Prevent sender from denying the action

Basic Cryptography Process



Components of Cryptography

01 Plaintext
The original readable message or data.
Example: Bank Password

02 Ciphertext
The encrypted unreadable form of plaintext.
Example: A7D9K2P

03 Encryption Algorithm
A mathematical process used to convert plaintext into ciphertext.
Examples: AES, DES, RSA

04 Decryption Algorithm
The process of converting ciphertext back into plaintext.

05 Key
A secret value used during encryption and decryption.
Types:
- Public Key
- Private Key
- Secret Key

06 Sender
The person or system sending encrypted data.

07 Receiver
The authorized person or system receiving and decrypting the data.

Types of Cryptography

1 Symmetric Key Cryptography

In this method, the same key is used for both encryption and decryption.

Plaintext → Secret Key → Ciphertext

Ciphertext → Same Secret Key → Plaintext

- **Features:** Fast, Simple, Suitable for large data
- **Advantages:** High speed, Less computational power
- **Disadvantages:** Key sharing is difficult, Less secure for large networks
- **Examples:** AES, DES, Blowfish
- **Real-Life Uses:** File encryption, Database protection, Wi-Fi security

2 Asymmetric Key Cryptography

Uses two different keys: **Public Key** and **Private Key**.

Public Key → Encryption

Private Key → Decryption

- **Features:** More secure, Solves key distribution problem
- **Advantages:** Better security, Supports digital signature
- **Disadvantages:** Slower than symmetric encryption, Requires more processing power
- **Examples:** RSA, ECC, Diffie-Hellman
- **Real-Life Uses:** SSL/TLS, Digital signatures, Secure email

3 Hash Functions (Hash Cryptography)

Converts data into a fixed-size hash value.

Data → Hash Function → #A7B3C9D2 (E.g. SHA-256)

- **Features:** One-way process, Cannot easily reverse original data
- **Advantages:** Ensures integrity, Fast verification
- **Disadvantages:** Cannot recover original message
- **Examples:** SHA-256, MD5, SHA-1
- **Real-Life Uses:** Password storage, Blockchain, Data integrity checking

Symmetric vs Assymetric

Symmetric vs Assymetric Cryptography

Feature	Symmetric	Asymmetric
Number of Keys	One key	Two keys
Speed	Faster	Slower
Security	Less secure	More secure
Key Sharing	Difficult	Easier
Examples	AES, DES	RSA, ECC

Applications of Cryptography

- Online Banking
- ATM Transactions
- E-commerce
- Secure Messaging
- Digital Signatures
- Blockchain
- Government Security Systems
- Password Protection

Common Cryptographic Attacks

01 Brute Force Attack

Trying all possible keys.

02 Man-in-the-Middle Attack

Attacker secretly intercepts communication.

03 Phishing

Tricking users to reveal secret information.

04 Dictionary Attack

Using common password lists to crack passwords.

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Classic Cryptography vs Modern Cryptography



Feature	Classic Cryptography	Modern Cryptography
Definition	Old traditional encryption techniques used in ancient times	Advanced computer-based encryption techniques used today
Technique	Mainly substitution and transposition methods	Mathematical algorithms and digital encryption
Key Usage	Usually single key	Single key or public-private key
Security Level	Less secure	Highly secure
Processing	Manual or simple mechanical process	Computerized and automated process
Speed	Slow	Fast and efficient
Complexity	Simple algorithms	Complex mathematical algorithms
Data Type	Mainly text messages	Text, audio, video, files, databases
Attack Resistance	Easy to break	Difficult to break
Examples	Caesar Cipher, Vigenère Cipher	AES, RSA, DES, ECC

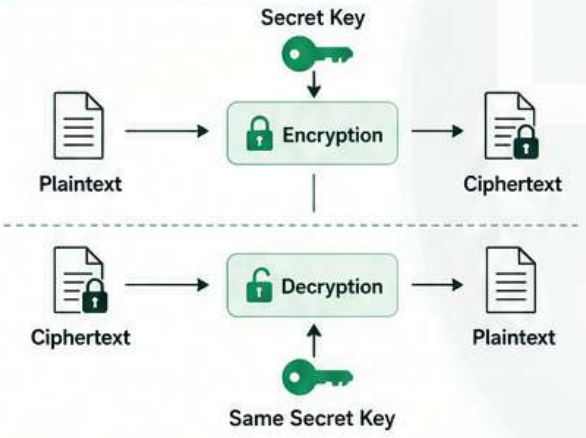
Symmetric Encryption

Symmetric Encryption is a cryptographic technique where the **same secret key** is used for both encryption and decryption.

It is also called:

- Secret Key Cryptography
- Private Key Cryptography

Block Diagram



Encryption Formula

$$C = E(P, K)$$

Where:

- P = Plaintext
- K = Secret Key
- C = Ciphertext

Decryption Formula

$$P = D(C, K)$$

How Symmetric Encryption Works

Step-by-Step Procedure

1. Sender writes plaintext
2. Plaintext is encrypted using a secret key
3. Ciphertext is transmitted
4. Receiver uses the same key to decrypt
5. Original plaintext is recovered

Methods (Types) of Symmetric Encryption

Block Cipher

Encrypts data in fixed-size blocks.

Features

- Data divided into blocks
- Each block encrypted separately

Examples

- AES
- DES
- Blowfish

Example



Stream Cipher

Encrypts data one bit or one byte at a time.

Features

- Continuous encryption
- Faster for streaming data

Examples

- RC4
- Salsa20

Example



Popular Symmetric Encryption Algorithms

Algorithm	Key Size	Features
DES	56-bit	Older and less secure
3DES	168-bit	Improved DES
AES	128/192/256-bit	Highly secure and modern
Blowfish	32–448-bit	Fast and flexible
RC4	Variable	Stream cipher

Advantages of Symmetric Encryption

1. High speed
2. Efficient for large data
3. Less computational power required
4. Simple implementation
5. Suitable for real-time communication

Disadvantages of Symmetric Encryption

1. Key distribution problem
2. Difficult key management in large networks
3. If key is leaked, security is broken
4. No direct support for digital signature
5. Less secure than asymmetric encryption for internet communication

Applications of Symmetric Encryption

- File encryption
- Database protection
- VPN security
- Wi-Fi security
- Cloud storage protection
- ATM transaction systems
- Banking systems

Real-Life Example

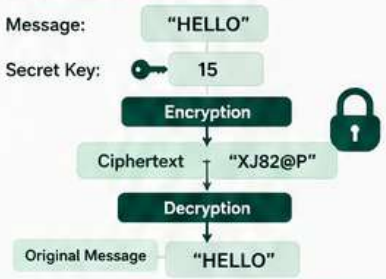
AES (Advanced Encryption Standard)

AES is the most widely used symmetric encryption algorithm.

Used in:

- Online banking
- WhatsApp encryption
- Government systems
- Military communication

Symmetric Encryption Flow Example



Asymmetric Encryption

Asymmetric Encryption is a cryptographic method where two different keys are used:

- Public Key → Used for encryption
- Private Key → Used for decryption

It is also called:

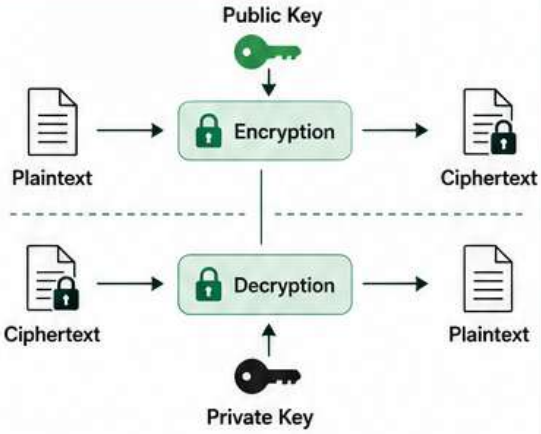
- Public Key Cryptography

Basic Concept

- Public Key → Encrypt Data
- Private Key → Decrypt Data

The public key can be shared openly, but the private key must remain secret.

Block Diagram



Encryption Formula

$$C = E(P, PU)$$

Where:

- P = Plaintext
- PU = Public Key
- C = Ciphertext

Decryption Formula

$$P = D(C, PR)$$

Where:

- C = Ciphertext
- PR = Private Key
- P = Plaintext

How Asymmetric Encrypetric Works

Step-by-Step Procedure

1. Receiver generates:
   - Public Key
   - Private Key
3. Receiver shares the public key publicly.
4. Sender encrypts the message using the receiver's public key.
5. Encrypted ciphertext is sent.
6. Receiver decrypts the ciphertext using the private key.

Types / Methods of Asymmetric Encryption

1 RSA (Rivest-Shamir-Adleman)

Most widely used asymmetric algorithm.

Features

- Uses large prime numbers
- Highly secure

Applications

- Secure web communication
- Digital signatures

2 ECC (Elliptic Curve Cryptography)

Features

- Smaller key size
- Faster than RSA
- Strong security

Applications

- Mobile devices
- IoT security

3 Diffie-Hellman Key Exchange

Features

- Used for secure key exchange
- Does not directly encrypt messages

Applications

- VPN
- SSL/TLS

Advantages of Asymmetric Encryption

1. Higher security
2. Solves key distribution problem
3. Supports digital signatures
4. Enables authentication
5. Suitable for internet communication

Disadvantages of Asymmetric Encryption

1. Slower than symmetric encryption
2. Requires more computational power
3. Complex implementation
4. Not efficient for large data encryption

Applications of Asymmetric Encryption

- Online banking
- SSL/TLS websites
- Secure email
- Digital signatures
- E-commerce security
- Cryptocurrency systems

Real-Life Example

RSA Algorithm

RSA is commonly used in:

- HTTPS websites
- Secure login systems
- Digital certificates

Working Example

Suppose:

- Public Key = Shared openly
- Private Key = Kept secret

Process

"HELLO" → Encrypt using Public Key → Ciphertext → Decrypt using Private Key → "HELLO"

Only the private key holder can read the original message.

List of Cryptographic Attack

A cryptographic attack is an attempt to break a cryptographic system, steal secret information, discover encryption keys, or access protected data without authorization.

Major Types of Cryptographic Attacks

	Attack Name	Main Target
1	Brute Force Attack	Encryption key
2	Dictionary Attack	Passwords
3	Ciphertext-Only Attack	Ciphertext
4	Known-Plaintext Attack	Plaintext-ciphertext pairs
5	Chosen-Plaintext Attack	Encryption system
6	Chosen-Ciphertext Attack	Decryption system
7	Man-in-the-Middle Attack	Communication channel
8	Replay Attack	Authentication process
9	Birthday Attack	Hash functions
10	Side-Channel Attack	Physical implementation
11	Phishing Attack	User credentials
12	SQL Injection	Database systems
13	Trojan Horse	Computer systems
14	Eavesdropping	Network communication

1 Brute Force Attack

Definition
An attacker tries all possible keys or passwords until the correct one is found.

Features
- Simple attack method
- Time-consuming

Example
Trying every possible ATM PIN combination.

Prevention
- Strong passwords
- Long encryption keys

2 Dictionary Attack

Definition
Uses a list of common passwords and dictionary words to crack passwords.

Example
Trying:

123456
password
admin

Prevention
- Use complex passwords
- Multi-factor authentication

3 Ciphertext-Only Attack (COA)

Definition
The attacker only has access to ciphertext and tries to determine plaintext or keys.

Target
Encrypted messages only.

A7D9K2P#Q9L

Prevention
- Strong encryption algorithms
- Randomized encryption

4 Known-Plaintext Attack (KPA)

HELLO → KHOOR

Definition
The attacker has:
- Plaintext
- Corresponding ciphertext

and uses them to discover the key.

Example
Knowing: HELLO → KHOOR

Prevention
- Advanced encryption algorithms
- Frequent key changes

5 Chosen-Plaintext Attack (CPA)

Definition
The attacker chooses plaintexts and observes the produced ciphertexts to break encryption.

Example
Testing encryption behavior with selected inputs.

Prevention
- Secure encryption modes
- Random padding

6 Chosen-Ciphertext Attack (CCA)

Definition
The attacker chooses ciphertexts and studies decrypted outputs.

Prevention
- Secure decryption protocols
- Authentication mechanisms

7 Man-in-the-Middle Attack (MITM)

Sender ↔ Receiver

Definition
An attacker secretly intercepts communication between two parties.

Example
Fake Wi-Fi hotspot stealing login information.

Prevention
- SSL/TLS
- VPN
- Authentication certificates

8 Replay Attack

Definition
An attacker captures valid transmitted data and retransmits it later.

Example
Reusing captured login session data.

Prevention
- Timestamps
- One-time passwords (OTP)

9 Birthday Attack

Definition
A cryptographic attack against hash functions using hash collisions.

Target
Hash algorithms like MD5 or SHA-1.

Prevention
- Strong hash functions
- SHA-256 or higher

10 Side-Channel Attack

Definition
Uses physical information from a system:
- Power consumption
- Timing information
- Electromagnetic leakage

Prevention
- Hardware protection
- Constant-time algorithms

11 Phishing Attack

Definition
Fake emails or websites trick users into revealing sensitive information.

Example
Fake bank login page.

Login
••••••

Prevention
- User awareness
- Secure websites

12 SQL Injection Attack

Definition
Malicious SQL commands are inserted into application inputs.

Example
Accessing database without authorization.

```
' OR '1'='1
```

Prevention
- Prepared statements
- Input validation

13 Trojan Horse Attack

Definition
Malicious software disguised as legitimate software.

Prevention
- Antivirus software
- Trusted downloads

14 Eavesdropping Attack

Definition
Secretly listening to private communication.

Prevention
- Encryption
- Secure communication channels